Tanmay Shukla

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Education

Georgia Institute of Technology | Atlanta, GA

Bachelor of Science in Computer Engineering, GPA 4.00

Expected Graduation: May 2027

Relevant Coursework: ECE 2020: Digital Systems Design**,** ECE 2031: Digital Design Lab, ECE 2035: Programming HW/SW Systems, ECE 2040: Circuit Analysis, CS 1331: Intro to Object Oriented Programming

Skills

RTL & Verification: FPGA Development (Intel Quartus, Xilinx Vivado), RTL (VHDL, Verilog/SystemVerilog), Synopsys VCS, Cadence Xcelium and Innovus, Universal Verification Methodology (UVM), Verdi, Static Timing Analysis, VLSI & Physical Design, TCL

Hardware Design & Embedded Systems: PCB Design & Layout (Altium, KiCad), High-Level Synthesis (HLS), ARM Mbed, STM32, Teensy, Raspberry Pi, Arduino, RISC-V Architecture, Computer Systems, Embedded Linux

Programming & Software Tools: C, C++, Python, Java, SQL, RISC V Assembly, Linux, Git/GitHub, SolidWorks, Bash/Shell scripting, Machine Learning (ML) models (GNNs), Pytorch and TensorFlow

Electronics, Testing & Communication Protocols: Circuit Analysis & Debugging, Oscilloscope, Logic Analyzer, UART, SPI, I^2C

Languages: English (Native), Hindi (Fluent), French (Basic)

Experience

FPGA Acceleration and High-Level Synthesis (HLS) Researcher – Sharc Lab, Georgia Tech *Aug 2025 – Present*

- Contributing to HLSFactory, an open-source HLS→RTL automation framework that enables AI-scale dataset generation for high-level chip design and uses ML models to optimize latency, power and resource utilization.
- Improved post-implementation Quality of Results (QoR) prediction accuracy of synthesizable HLS code by over 20% by implementing design space expansion and cross vendor synthesis across toolchains (Xilinx, Intel).

Design Verification Engineer, SiliconJackets *Aug 2025 - Present*

- Developing SystemVerilog testbench for Reorder Buffer (ROB) in an out-of-order RISC-V processor, verifying circular buffer allocation, result forwarding, and in-order retirement across parameterized fire/retire widths for silicon tapeout.
- Designed and verified a 64-bit calculator in SystemVerilog RTL with FSM-based control, synchronous resets, and a dual-bank SRAM interface. Completed synthesis, floorplanning, placement, and routing in Cadence Innovus using TCL to produce a full physical design layout (VLSI).

Avionics Hardware Lead, Stack Systems Engineer, Georgia Tech Experimental Rocketry *Jan 2025 – Present*

- Led avionics hardware team and collaborated with other subteams in designing the architecture and fabricating custom PC104 embedded systems stack for the 1st collegiate two-stage rocket targeting the Kármán line (Space).
- Designed and routed the ADCS PCB board (flight computer) for flight control and telemetry, projected to reach 100km.
- Co-developed hardware/software architecture, including 1000+ lines of embedded C and embedded linux code for control algorithms.

Projects

Four Channel Servo PWM Controller in VHDL (FPGA) *May 2025 – July 2025*

- Built a hardware PWM controller in VHDL that lets a soft-core processor drive four hobby servos with 1.4° resolution and 0 timing jitter fully autonomous operation on a DE10-Lite FPGA.

Missile Blaster – Arcade Shooter on ARM Mbed (C/C++) *May 2025 – August 2025*

- Developed an embedded arcade-style shooter on the ARM Mbed microcontroller platform in C/C++, interfacing with a color LCD for graphics, input controls, and audio peripherals to implement responsive gameplay features.

Independent Research on Graphene Supercapacitors and its efficiency over traditional EDLCs *June 2022 – December 2023*

- The Graphene Supercapacitor demonstrated a 40% higher energy density compared to an EDLC of the same parameters.
- Presented this research at a national science fair and won 1st place out of 3000 Students and 20+ schools.

PID line-following robot that doubled as a Bluetooth controlled Ps5 robot *Nov 2023-July 2024*

- Won first place at a robotics competition at Modern College out of 500+ students in schools & universities across Oman.
- Designed the robot chassis using CAD and used IR sensors for line-following and an HC05 module to control the robot using a Ps5 Controller through Bluetooth.